

PHYS 230 - Electricity and Magnetism

Lecture 34

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

9 Electromagnetic Induction

- Faraday's Law
- Lenz's Law

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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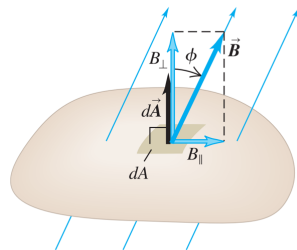
Magnetic Field Flux

As usual we define the infinitesimal flux $d\Phi_B$ of the magnetic field $\mathbf{B}$ through an infinitesimal surface $d\mathbf{A}$ as

$$d\Phi_B = \mathbf{B} \cdot d\mathbf{A}$$

where

Figure 29.3 Calculating the magnetic flux through an area element.



Magnetic flux through element of area $d\vec{A}$:

$$d\Phi_B = \vec{B} \cdot d\vec{A} = B_{\perp} dA = B dA \cos \phi$$

Magnetic Field Flux

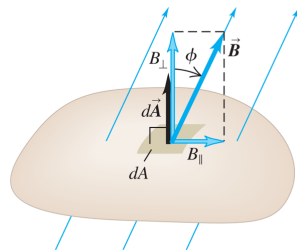
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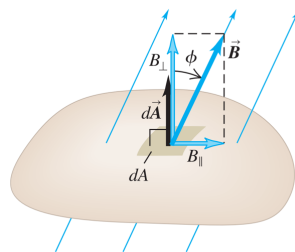
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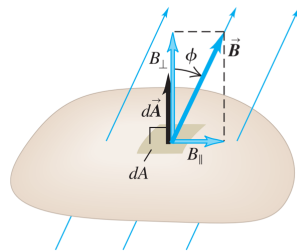
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The total flux through the finite area is

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

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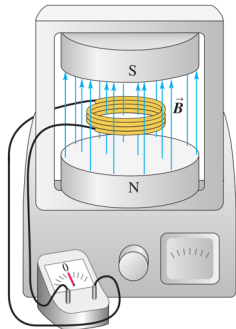
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Induced Current

Long time ago, experiments revealed that if one changes the flux

$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$ through a **closed** conducting **loop**, there will be a **current induced in the closed loop!**

Figure 29.2 A coil in a magnetic field. When the $\vec{B}$ field is constant and the shape, location, and orientation of the coil do not change, no current is induced in the coil. A current is induced when any of these factors change.



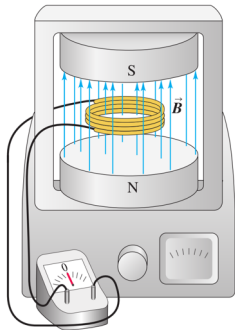
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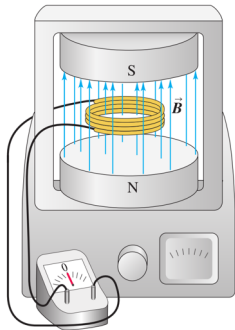
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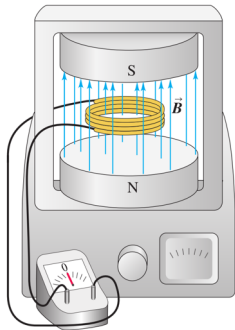
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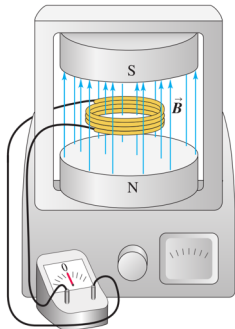
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- Change in $\mathbf{B}$ (magnitude, direction, or both)
- Change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both)
- Change in $\mathbf{B} \cdot d\mathbf{A}$, i.e., the orientation of $\mathbf{B}$ and $d\mathbf{A}$ with respect to each other

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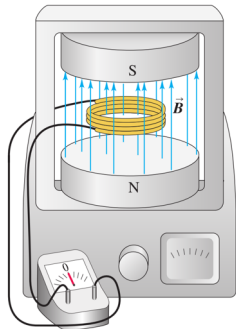


Faraday's Law

Faraday understood that the magnitude of the emf induced in a conducting loop due to change of Φ_B through it is

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

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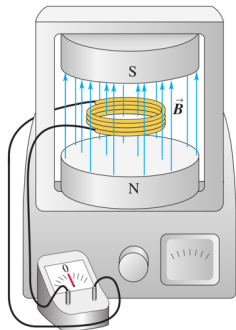
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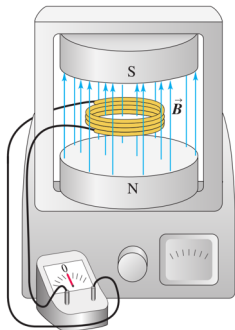
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What is the negative sign in the formula? and how do we find out the direction of $\mathcal{E}$?

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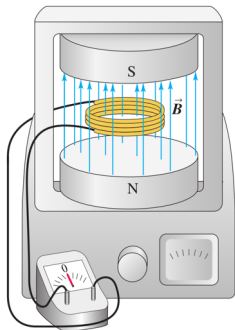
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This is the subject of Lenz's law...

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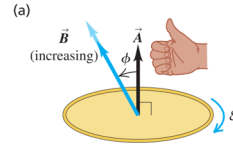
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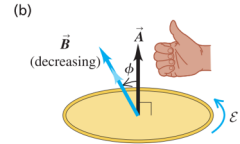
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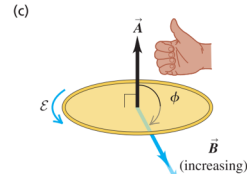
The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.



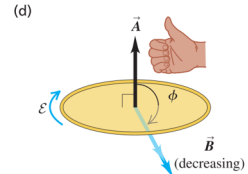
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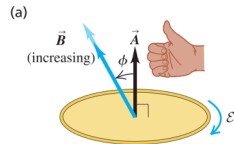


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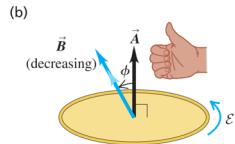
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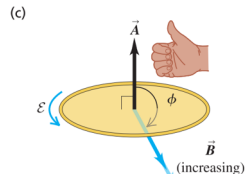
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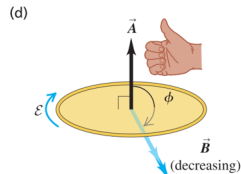
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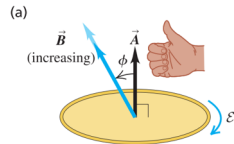


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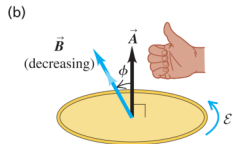
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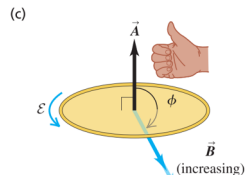
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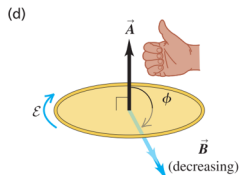
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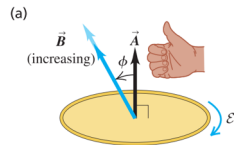


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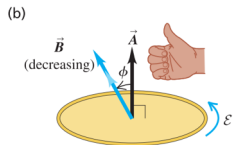
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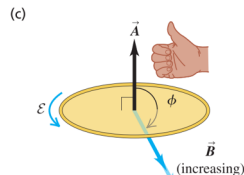
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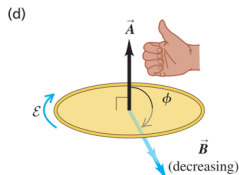
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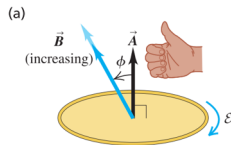


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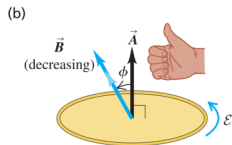
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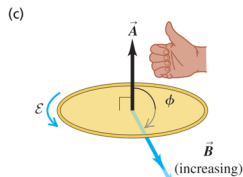
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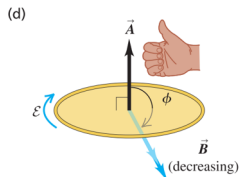
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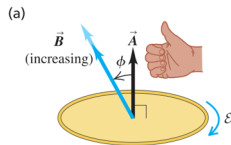


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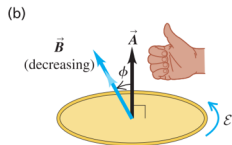
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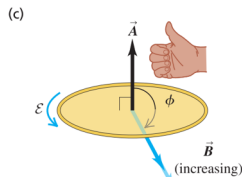
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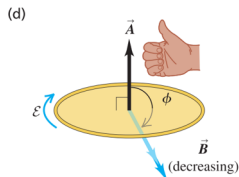
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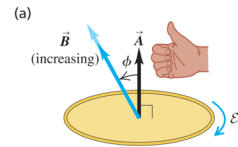
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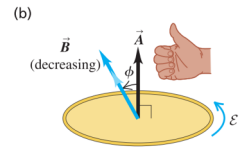
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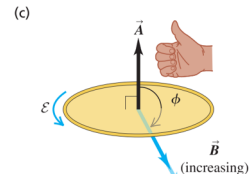
How to find the direction and sign of induced $\mathcal{E}$ and I :



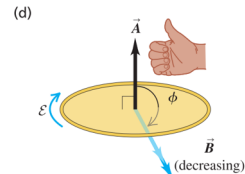
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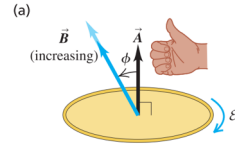


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

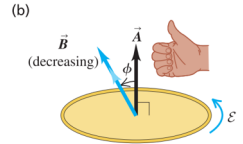
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

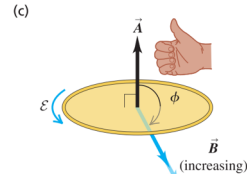
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:



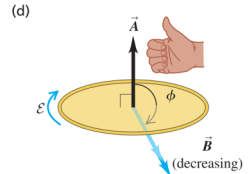
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



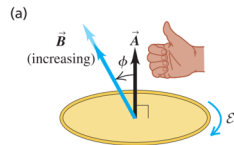
- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

Lenz's Law

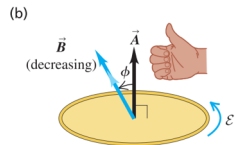
How to find the direction and sign of induced $\mathcal{E}$ and I :

I. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:

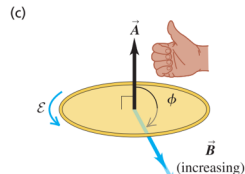
I.I $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)



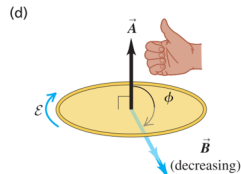
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
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- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

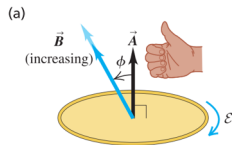
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

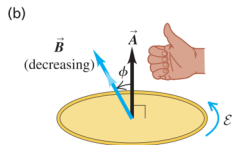
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:

1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)

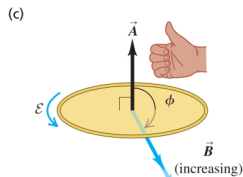
1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B



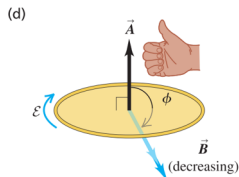
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
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- ... and becoming more negative ($d\Phi_B/dt < 0$).
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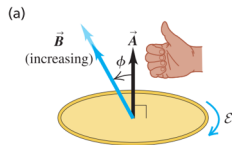


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

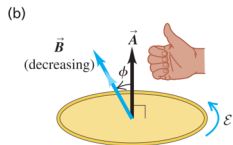
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

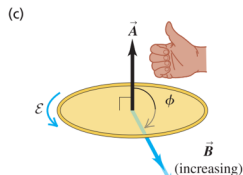
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:
 - 1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)
 - 1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B
2. If $|\Phi_B|$ is increasing/decreasing, the direction of I is such that the new $\mathbf{B}'$ it creates, wants to stop $|\Phi_B|$ from increasing/decreasing



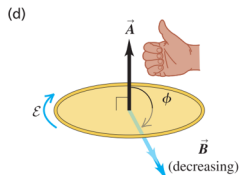
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



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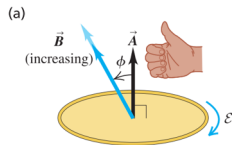


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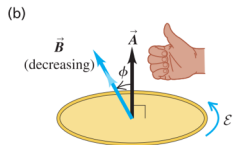
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

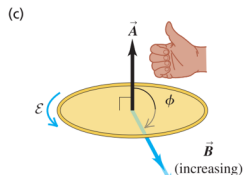
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:
 - 1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)
 - 1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B
2. If $|\Phi_B|$ is increasing/decreasing, the direction of I is such that the new $\mathbf{B}'$ it creates, wants to stop $|\Phi_B|$ from increasing/decreasing
3. The direction of $\mathcal{E}$ is the same as I



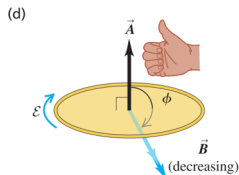
- Flux is positive ($\Phi_B > 0$) ...
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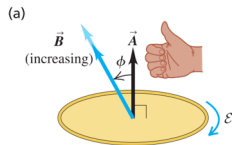


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

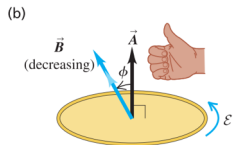
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

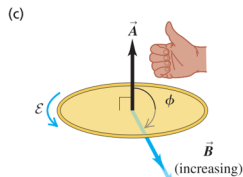
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:
 - 1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)
 - 1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B
2. If $|\Phi_B|$ is increasing/decreasing, the direction of I is such that the new $\mathbf{B}'$ it creates, wants to stop $|\Phi_B|$ from increasing/decreasing
3. The direction of $\mathcal{E}$ is the same as I
4. The sign of $\mathcal{E}$ is positive if $\mathbf{B}'$ is the direction of $\mathbf{A}$, negative if $\mathbf{B}'$ is the opposite direction of $\mathbf{A}$



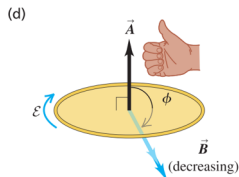
- Flux is positive ($\Phi_B > 0$) ...
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- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
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- Flux is negative ($\Phi_B < 0$) ...
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- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

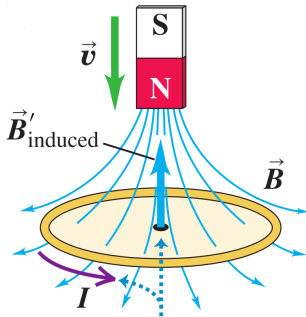
Based on change in Φ_B :

Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):

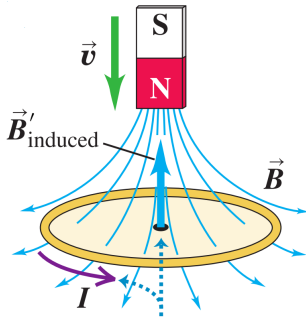


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$

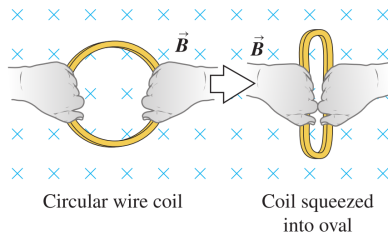


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):

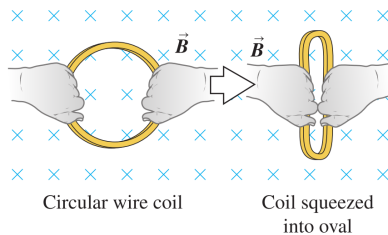


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{A}$

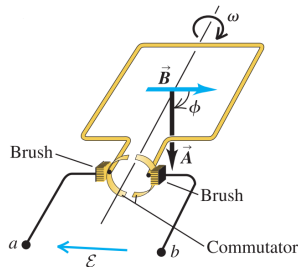


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{A}$
- If change in $|\Phi_B|$ is due to a change in $\mathbf{B} \cdot d\mathbf{A}$, i.e., the orientation of $\mathbf{B}$ and $d\mathbf{A}$ with respect to each other

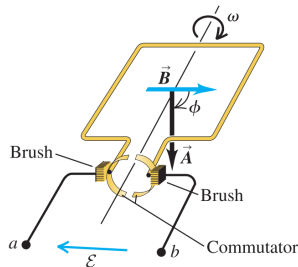


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

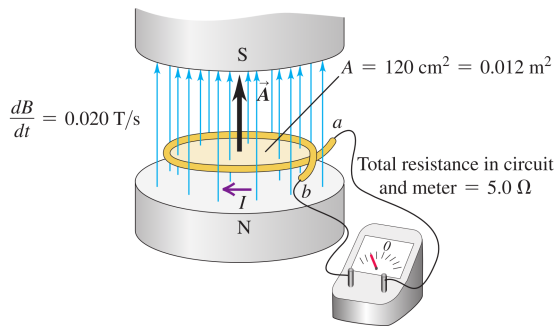
- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{A}$
- If change in $|\Phi_B|$ is due to a change in $\mathbf{B} \cdot d\mathbf{A}$, i.e., the orientation of $\mathbf{B}$ and $d\mathbf{A}$ with respect to each other
 - I will be such that induced $\mathbf{B}'$ stops further change in the angle between $\mathbf{B}$ and $\mathbf{A}$



Exercise 1

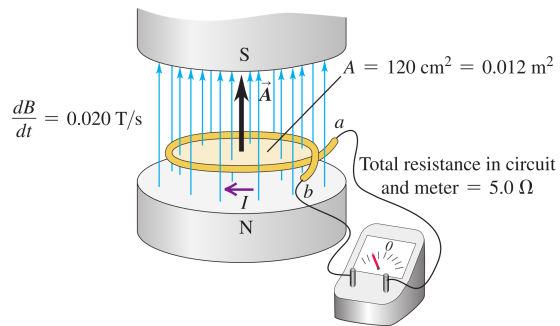
<https://epoll.srv.ualberta.ca> – Code: MBD

The magnetic field between the poles of the electromagnet in figure is uniform at any time, but its magnitude is increasing at the rate of 0.020 T/s . The area of the conducting loop in the field is 120 cm^2 , and the total circuit resistance, including the meter, is 5.0Ω . (a) Find the induced emf and the induced current in the circuit. (b) If the loop is replaced by one made of an insulator, what effect does this have on the induced emf and induced current?



Exercise 1

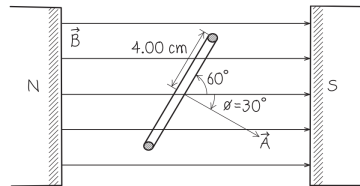
<https://epoll.srv.ualberta.ca> – Code: MBD



Exercise 2

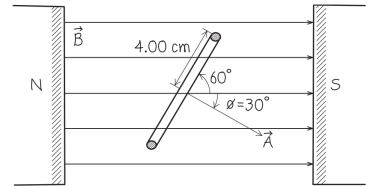
<https://epoll.srv.ualberta.ca> – Code: MBD

A 500-loop circular wire coil with radius 4.00 cm is placed between the poles of a large electromagnet. The magnetic field is uniform and makes an angle of $\phi = 60^\circ$ with the plane of the coil; it decreases at 0.200 T/s. What are the magnitude and direction of the induced emf?



Exercise 2

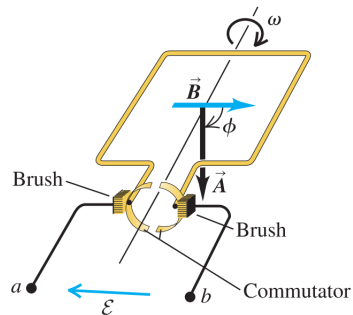
<https://epoll.srv.ualberta.ca> – Code: MBD



Exercise 3

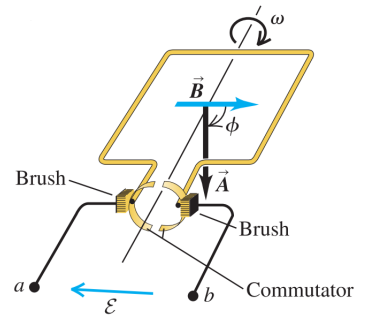
<https://epoll.srv.ualberta.ca> – Code: MBD

Figure shows a simple alternator, a device that generates an emf. A rectangular loop is rotated with constant angular speed ω about the axis shown. The magnetic field $\mathbf{B}$ is uniform and constant. At time $t = 0$, we have $\phi = 0$. Determine the induced emf.



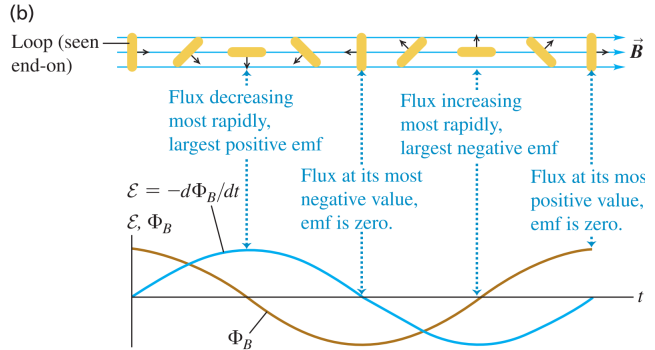
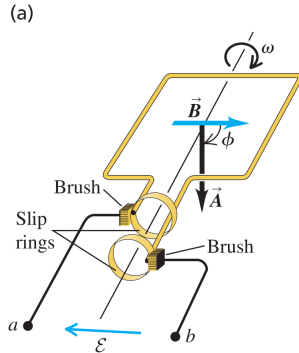
Exercise 3

<https://epoll.srv.ualberta.ca> – Code: MBD



Exercise 3

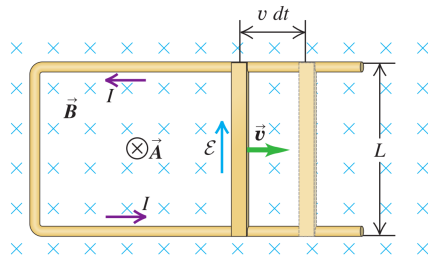
<https://epoll.srv.ualberta.ca> – Code: MBD



Exercise 4

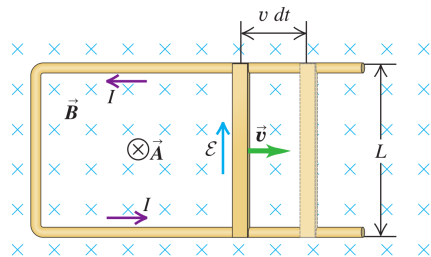
<https://epoll.srv.ualberta.ca> – Code: MBD

Figure shows a U-shaped conductor in a uniform magnetic field $\mathbf{B}$ perpendicular to the plane of the figure and directed into the page. We lay a metal rod (the “slidewire”) with length L across the two arms of the conductor, forming a circuit, and move it to the right with constant velocity $\mathbf{v}$. This induces an emf and a current, which is why this device is called a slidewire generator. Find the magnitude and direction of the resulting induced emf.



Exercise 4

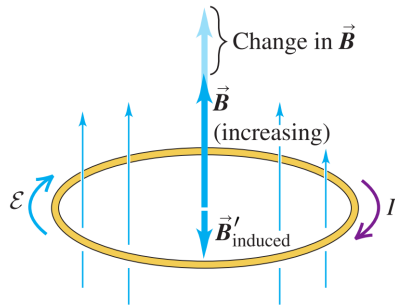
<https://epoll.srv.ualberta.ca> – Code: MBD



Exercise 5

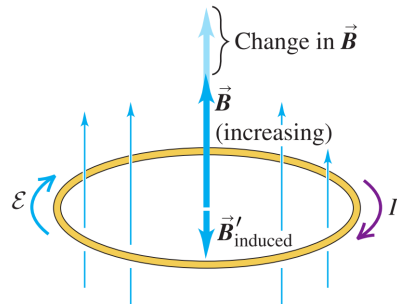
<https://epoll.srv.ualberta.ca> – Code: MBD

In figure the magnitude of the magnetic field $\vec{B}$ through the coil is increasing, so there is an induced emf. Use Lenz's law to determine the direction of the resulting induced current.



Exercise 5

<https://epoll.srv.ualberta.ca> – Code: MBD

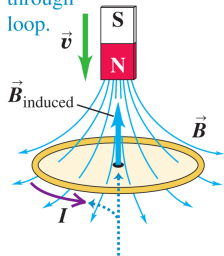


Exercise 5

<https://epoll.srv.ualberta.ca> – Code: MBD

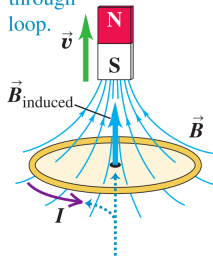
Figure 29.14 Directions of induced currents as a bar magnet moves along the axis of a conducting loop. If the bar magnet is stationary, there is no induced current.

- (a) Motion of magnet causes *increasing downward flux* through loop.

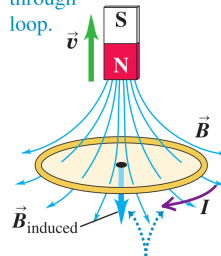


The induced magnetic field is *upward* to oppose the flux change. To produce this induced field, the induced current must be *counterclockwise* as seen from above the loop.

- (b) Motion of magnet causes *decreasing upward flux* through loop.



- (c) Motion of magnet causes *decreasing downward flux* through loop.



The induced magnetic field is *downward* to oppose the flux change. To produce this induced field, the induced current must be *clockwise* as seen from above the loop.

- (d) Motion of magnet causes *increasing upward flux* through loop.

